

Sub-Nanomolar Detection and Discrimination of Microcystin Congeners Using Aerolysin Nanopores

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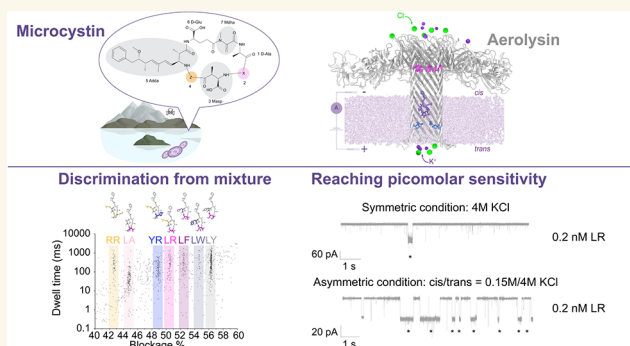
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ABSTRACT: Climate-driven disruptions in aquatic ecosystems are amplifying cyanotoxin production, threatening drinking and recreational water safety. Monitoring of these toxins is challenged by requirements of the low $\mu\text{g/L}$ detection limits and structural diversity. Here, we employ aerolysin nanopores to distinguish seven of the most prevalent microcystin congeners, both individually and in mixtures, at environmentally relevant concentrations. Importantly, we showed that aerolysin enables the detection of microcystins in spiked and real contaminated lake water samples at concentrations below the World Health Organization's intervention thresholds, reaching picomolar sensitivity. Moreover, combining experiments and molecular dynamics simulations, we further investigated the microcystin sensing mechanism, suggesting that the ionic current blockade is primarily governed by K238 in aerolysin, while dwell time is regulated by the R220 constriction site. Our results support the use of nanopore sensing technology for real-time monitoring of microcystins in drinking water sources and surface waters.

KEYWORDS: nanopore sensing, label-free biosensing, single-molecule analysis, real-time monitoring, environmental monitoring, toxic cyanobacterial blooms, water quality



INTRODUCTION

Aquatic ecosystems and water quality are increasingly threatened by both anthropogenic pollutants and natural toxins, particularly those produced by cyanobacteria.¹ Climate change, eutrophication, and rising CO₂ levels have led to more frequent cyanobacterial blooms,^{2–4} which negatively impact biodiversity² and pose significant risks to human health.⁵ During these blooms, cyanobacteria release a range of potent cyanotoxins—including neurotoxins, cytotoxins, and hepatotoxins such as microcystins, nodularins, and cylindrospermopsin—that inhibit key metabolic enzymes.^{6–9} Cyanotoxins, especially microcystins, can accumulate to micromolar concentrations in surface scums¹⁰ and nanomolar levels in the water column,¹¹ well above the 1 $\mu\text{g/L}$ (equivalent to 1 nM) safety guideline for chronic exposure to drinking water set by the World Health Organization (WHO).¹² Even at lower concentrations, these toxins may persist before and after blooms in meso-oligotrophic lakes like Lake Geneva (Switzer-

land),¹³ impacting aquatic life despite being below regulatory thresholds.

Microcystins (MCs)^{14,15} are cyclic peptides made of canonical and three noncanonical amino acids: D-erythro- β -methyl-isopartate (D-Masp), 3-amino-9-methoxy-2,6,8-trimethyl-10-phenyldeca-4,6-dienoic acid (Adda), and N-methyldehydroalanine (Mdha) for the congener MC-LR (Figure 1A, located at positions 3, 5, and 7, respectively). Variations of this general scaffold can produce more than 300 congeners identified to date.^{16,17} MCs are well-documented public health hazards; for example, a 1996 cyanobacterial bloom in Brazil led to the deaths of over 50 dialysis patients

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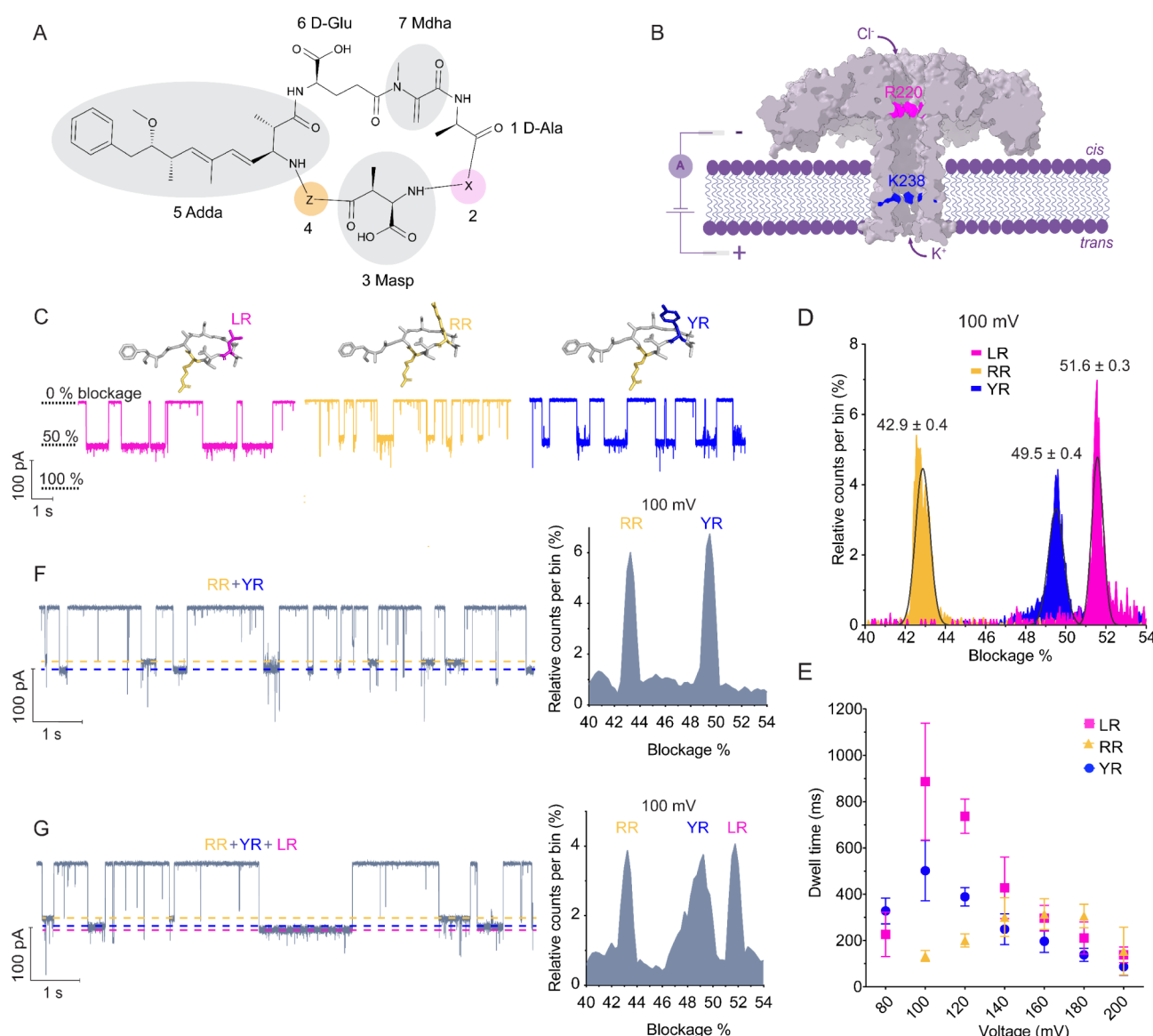


Figure 1. Detection of MC-LR, MC-RR, MC-YR with WT AeL nanopore. (A) Microcystin 2D structure with its noncanonical residues (gray) and 2 variable regions: X (pink) and Z (yellow). (B) Cartoon of the WT AeL pore (PDB: 9FM6)⁵¹ with two narrowest constrictions: R220 (magenta) and K238 (blue). Membrane (purple), MC addition to the *cis* chamber (not to scale), and passage of potassium and chloride ions are shown. (C) Typical traces recorded at 100 mV in 4 M KCl buffered with 10 mM Tris and 1.0 mM EDTA at pH 7.5. Variations in blockage signatures can be distinguished among the 3 congeners (PDB of MC-LR: 1LCM;⁷⁵ RR and YR were generated from 1LCM in PyMOL⁷⁶). Selected 10 s fragments of the current traces, recorded at a sampling rate of 50 kHz and filtered with a 500 Hz low-pass filter, are shown. (D) Histograms of current blockage produced by MCs at 100 mV in individual experiments (with at least $n = 3$ separate experiments per congener), showing the current separation between the three different MCs. (E) Dwell time dependence on voltage. Each point is the result of the average of at least $n = 3$ pore replicates (Figures S1–S3, S5–S8). (F, G) Raw current traces (recorded at 50 kHz, filtered to 500 Hz) and histograms (bin size = 0.25) showing blockage events of MCs and their distribution in a mixture of either two or three congeners.

due to acute liver failure.^{13,18} This incident highlighted the need for effective monitoring of MC concentrations in water. Human exposure typically occurs via oral or respiratory routes,¹³ with symptoms ranging from mild skin rashes and nausea to severe gastrointestinal damage, liver damage, and respiratory distress. At the molecular level, MCs are actively taken up by hepatocytes and potentially inhibit the protein phosphatases PP1 and PP2A, disrupting cellular signaling and cytoskeletal integrity,^{19,20} which can lead to cell death and, in severe cases, fatal liver failure.²¹ The use of contaminated water for irrigation also raises concerns for food safety.²² In response to these risks, the WHO established three guideline values

(Table S1) that trigger specific interventions,¹² with the guideline for drinking water set at 1 $\mu\text{g/L}$ (equivalent to 1 nM) for the best-studied microcystin congener, MC-LR.²²

To date, MCs are commonly monitored by enzyme-linked immunosorbent assay (ELISA, US EPA Method 546²³), which is easy to operate and enables rapid, picomolar-range²⁴ detection. However, ELISA lacks specificity for individual MC congeners^{25,26} and may cross-react with similar toxins,^{27,28} potentially causing false positives. In contrast, liquid chromatography coupled to mass spectrometry (LC–MS/MS, US EPA Method 544²⁹) delivers the precise, sensitive, and selective analysis of 11 MC congeners for which reference

standards are available,³⁰ reaching sensitivities in the low nanomolar range (compatible with environmental samples³¹). However, the high cost of equipment, the need for qualified users, sample transportation, and the long turnaround time until results are available make this method expensive and impractical for real-time analysis.²⁹ Currently, LC–MS/MS analysis of MCs takes from 1 day for quantitative results to several weeks for detailed suspect screening where no reference standards are available.³² This delay limits timely interventions and increases the risk of missing sudden toxin spikes, as cyanotoxin levels can rapidly exceed safe limits due to blooms or cell lysis.³³ Real-time monitoring is therefore essential for promptly detecting toxin peaks and enabling timely responses to protect drinking and recreational water—especially for vulnerable groups such as children^{34,35} and pets.^{36–39}

Other detection methods include fluorescence, spectroscopy, and biosensors^{40–48} such as biological nanopores.^{49,50} Nanopore sensing allows single-molecule detection by monitoring ionic current fluctuations as molecules interact with a biological pore. Recent advances in nanopore research have shown that three of the most abundant MC congeners (MC-LR, -YR, and -RR) can be distinguished by their unique current signatures using the α -hemolysin (α -HL)^{49,50} nanopore. However, these measurements required relatively high concentrations (1 μ M,⁵⁰ 200 nM⁴⁹), well above the WHO safety threshold.

Here, we present a nanopore-based approach for the simultaneous detection of MCs at concentrations below the second strictest WHO threshold (<12 nM, Table S1) by using wild-type aerolysin (WT AeL). WT AeL is a β -barrel pore-forming toxin (β -PFT) characterized by a narrow constriction (1 nm in diameter) and an extended channel length (10 nm). Its structure was recently solved in a native lipid environment at high resolution⁵¹ and has shown remarkable sensitivity and prolonged analyte residence time compared to other β -PFTs. Aerolysin sensing properties have been extensively studied,^{52,53} and various engineered mutants have enabled enhanced analysis of synthetic polymers,⁵⁴ short single-stranded DNA oligomers,⁵⁵ 13 single amino acid substitutions within a short polypeptide,⁵⁶ discrimination of native proteins with similar molecular mass based on the current blockage fingerprints generated by the respective trypsin-digested polypeptide fragments,⁵⁷ and other single-molecule targets.⁵⁸ These include applications such as temporal omics,⁵⁹ peptide biomarker detection,^{60,61} differentiation of oxytocin and its structural variants,⁶² characterization of isomerically diverse ginsenosides,⁶³ stereoisomer identification within peptides,⁶⁴ identification of highly relevant post-translational modifications (PTMs),^{65–68} screening the crosstalk effects between angiotensin-converting enzymes,⁶⁹ and studying catalytic heterogeneity of enzymatic reactions.⁷⁰ Using WT AeL, this study achieved picomolar detection of MC-LR at levels well below the strictest WHO threshold (<1 nM) by using an asymmetric salt gradient. We furthermore showed the capability of discriminating MC congeners while in mixture in a lake water background at an environmentally relevant concentration, and we successfully detected and quantified MC-LR in a real contaminated lake sample. Moreover, combining experiments and molecular dynamics simulations, we identified how single-point mutations influence the sensitivity of aerolysin, shedding light on its mechanism for microcystin discrimination. Our results set the basis to use

biological nanopores for real-time detection and monitoring of the presence and abundance of microcystins in rivers and lakes.

RESULTS

Characterization of MC-LR, MC-RR, and MC-YR with Wild-Type Aerolysin Nanopore

Single-channel recordings using WT AeL nanopore were conducted to investigate current blockage characteristics and assess the translocation of MCs. The WT AeL transmembrane barrel is formed by the oligomerization of seven identical protomers into a heptameric pore.^{52,71,72} The experimental setup (Figure 1B) comprised *cis* and *trans* electrolyte-filled chambers, separated by a membrane and connected by a single nanopore. The key constriction sites, R220 and K238, are known to be critical for sensing applications.^{55,56,61,68,73} Upon applying an electrical bias, a stable ionic current (open pore current, I_0) is generated. Introduction of an analyte into the *cis* chamber induces transient current blockades, characterized by their magnitude (I), dwell time (the duration the analyte spends in the pore), and event frequency—parameters that provide insights into the analyte's molecular structure and translocation dynamics.

The selected MCs exhibit structural variability at positions 2 (X) and 4 (Z), as shown in Figure 1A. The 3 most studied congeners of MCs differ by a single residue at the X site, where either a leucine, arginine, or tyrosine (L, R, Y) residue can be found. Initially, we focused on congeners that solely differ at the X site while maintaining a fixed arginine residue at site Z. These congeners, denoted as MC-LR, MC-RR, and MC-YR, exhibit net charges of -1 , 0 , and -1 at pH 7.5, respectively.

Experimental conditions were adapted from previous studies,^{50,74} using 4 M KCl to enhance capture and translocation of neutral, negatively, and positively charged peptides through WT AeL by increasing electro-osmotic flow (EOF) and reducing electrostatic interactions.⁷⁴ This approach enabled sensitive detection of MCs at 50 nM—substantially lower concentrations than previously used in the α -HL nanopore studies.^{49,50}

As illustrated in Figure 1C, 50 nM concentrations of MCs yielded clear and prolonged blockade events (Figures S1–S3). Current blockage and dwell time served as the primary parameters for congener differentiation. Mean dwell times were calculated from exponential fits of dwell time distributions. Scatterplots of dwell time vs current blockage (Figures S5–S8) were generated across voltages from 60 to 200 mV to determine translocation thresholds, inferred from decreasing dwell times with increasing voltage. A voltage of 100 mV provided optimal separation among the 3 congeners (Figure 1D), with MC-RR showing 6.6% and 8.7% lower current blockage than -YR and -LR, respectively.

MC current blockage followed the trend: MC-RR < -YR < -LR, with MC-YR and -LR differing by only 2.1%. Dwell time analysis (Figure 1E) showed that MC-LR and -YR, both negatively charged, began translocating at lower voltages compared to neutral -RR. The longest dwell time for MC-RR (314.2 ± 66.4 ms) occurred at 160 mV, indicating that higher voltage is needed to drive its translocation due to its neutral charge.

Although larger molecular volume is often associated with greater current blockage, no systematic correlation was found between the volume of the variable amino acid and the respective congener's current blockage (Table S2). Despite

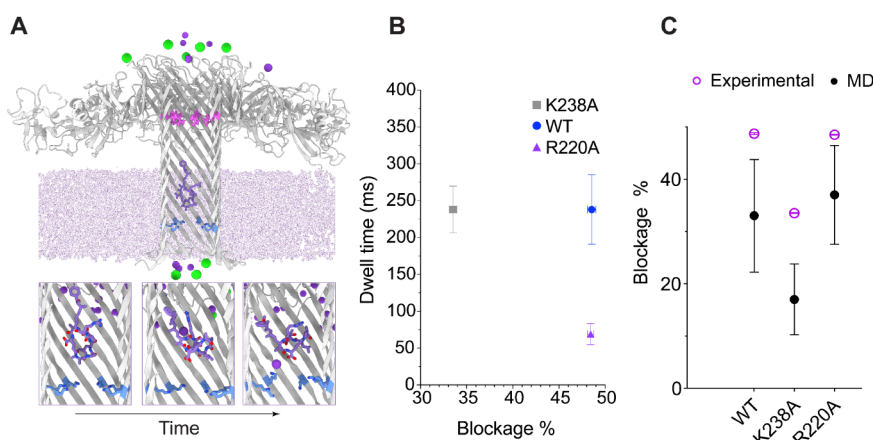


Figure 2. Dissecting MC-LR blockage with mutagenesis and MD simulations. (A) MD setup (top) and changes in MC-LR orientation within the pore over time (bottom). Cartoon of WT AeL with highlighted R220 (magenta) and K238 (blue) constrictions; MC-LR structure (purple); K^+ , Cl^- ions (dark purple and green spheres, respectively) are visualized. Cartoon of WT AeL showing R220 (magenta) and K238 (blue) constrictions, MC-LR (purple) placed inside the β -barrel, and visualized K^+ (dark purple) and Cl^- (green) ions. As demonstrated by the frames of MC-LR from the beginning and end of the simulation run, the orientation and conformation of -LR clearly change. (B) MC-LR experimental dwell time and blockage comparison between WT AeL ($48.5 \pm 0.4\%$), K238A ($33.5 \pm 0.4\%$), and R220A ($48.4 \pm 0.3\%$) mutants at 160 mV (at least $n = 3$ pore replicates for each mutant were analyzed). Experimental data acquired in 4 M KCl, buffered with 10 mM Tris and 1.0 mM EDTA at pH 7.5. (C) MC-LR experimental and MD predicted blockage comparison between AeL mutants at 160 mV. Experimental blank traces of AeL pores and IV curves are shown in Figure S13, and MC-LR event detection in K238A and R220A is shown in Figure S14.

arginine's (R) larger volume ($173.4 \text{ \AA}^3$) compared to leucine (L) ($166.7 \text{ \AA}^3$),⁷⁷ MC-RR generated less blockage than -LR, consistent with prior α -HL data. Similarly, MC-YR, with a larger tyrosine residue (Y) ($193.6 \text{ \AA}^3$), showed less blockage than -LR. This suggests that factors beyond volume—such as charge, conformation, and hydrophobicity—influence blockage behavior. MC-RR showed the smallest blockage, likely due to its +1 charge, compared to -LR and -YR, promoting intramolecular interactions and reducing the interaction with AeL (also reflected in Figure 1E by short dwell times), likely resulting in a more compact conformation in the long and narrow β -barrel of AeL. To determine how the cyclic structure of MCs affects current blockage, we synthesized linear versions of these three MCs, where non-natural residues were replaced with similar-sized and -charged natural amino acids, preserving overall the net charge relative to each cyclic analogue (Figure S10A,B). The linear MCs produced significantly faster dwell times and broader blockage distributions (Figure S10C,D, Figure S11), underscoring that both the bulkier structure of cyclic MCs and the presence of non-natural amino acids contribute to providing improved sensing performance.

The increasing blockage from MC-RR to -YR and -LR aligns with their increasing negative charge and hydrophobicity. RR, being neutral and least hydrophobic, interacts weakly with the pore, whereas LR, the most hydrophobic and negatively charged, exhibits stronger binding and longer dwell times. YR shows intermediate behavior, consistent with its physicochemical profile.

NMR spectroscopy was used to probe the conformational states of the 3 MCs in solution by comparing their backbone 1H and ^{13}C chemical shifts, which are exquisitely sensitive to conformation (Table S3). These data show that MC-LR and -YR must be structurally more similar to each other than to -RR (Figure S9). In particular, the $^{13}C\alpha$ and $^{13}C\beta$ chemical shifts at E6 differ markedly between MC-LR and MC-RR (^{13}C -HSQC spectra were not attainable for MC-YR due to its low concentrations, but 1H -TOCSY patterns are almost identical to

MC-LR's). Although explicit interpretation of the chemical shift changes is not trivial, the higher $\Delta\delta_{C\alpha} - \Delta\delta_{C\beta}$ in MC-RR would indicate a slightly higher level of local structure. Molecular dynamics simulations for MCs in solution⁷⁸ have shown that these are very flexible molecules, and it appears from these NMR data that the conformational landscape is altered in MC-RR compared to MC-LR and MC-YR. These differences would be consistent with the trends found in current blockages (Figure 1D), and if it were the root cause of the observed trends, it would imply overall somewhat tighter states in MC-RR. However, what was not observed in the NMR spectra—but becomes evident in the nanopore experiments between 140 mV and 200 mV (Figures 3A, S6, S7)—is the presence of two distinct populations for MC-RR, suggesting that the congener may adopt multiple orientations and conformations upon entering and interacting with the pore, as similarly observed in MD simulations of MCs in solution.⁷⁸

Finally, a mixture experiment tested whether the WT AeL could resolve all three congeners simultaneously. An initial binary mixture of MC-RR and -YR (Figure 1F) was followed by -LR addition (Figure 1G) at adjusted concentrations (25 nM MC-LR, 50 nM -RR, 20 nM -YR) to balance capture rates to ensure uniform signal distribution (Figure S4). The resulting current blockage distributions aligned well with the individual experiments, confirming that WT AeL can distinguish MC congeners that differ by a single residue. The capture frequency of the negatively charged congeners MC-LR and -YR increased more rapidly with increasing voltage compared to neutral -RR (Figure S4). This might be due to the combined effect of electrophoretic mobility and electroosmotic flow (EOF), leading to an increased frequency of negatively charged species at higher voltages, whereas neutral species are less affected by electrophoretic force.^{56,79}

Dissecting the Microcystin Translocation Mechanism

To gain more insights into which region of the pore influences the current blockage the most, we compared results from

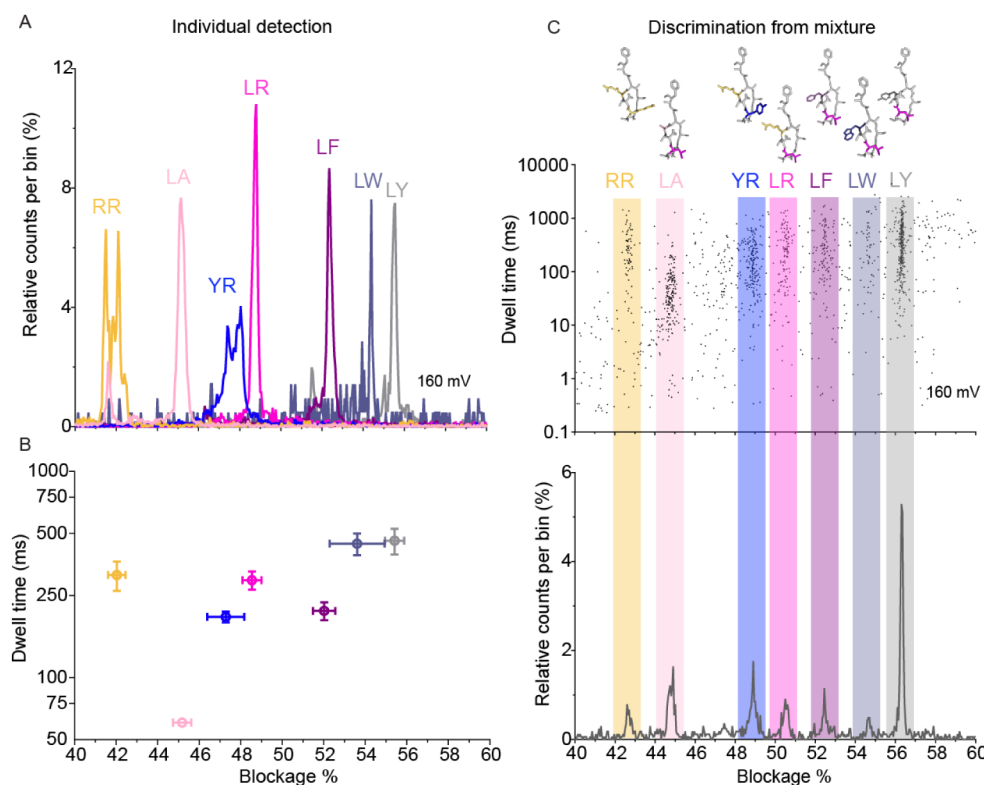


Figure 3. Expanding detection of MCs with WT AeL. (A) Each MC congener produces a distinguishable current blockage in individual experiments at nanomolar concentration at 160 mV, represented in a histogram ($n > 500$ events). (B) Discrimination of 7 congeners at nanomolar concentrations in individual experiments, based on current blockage and dwell time at 160 mV. Each point is the result of the average of at least $n = 3$ pore replicates. (C) Representation of 7 MCs, a scatterplot derived from a single nanopore recording of a mixture containing the 7 MCs (top), and a resulting histogram (bottom) are shown. Data acquired in 4 M KCl, buffered with 10 mM Tris and 1.0 mM EDTA at pH 7.5. Conserved leucine is indicated in pink, and the variable amino acid is colored according to the histogram. Raw current traces (Figure S27) and molecular properties of MCs are shown in Table S2.

nanopore experiments and molecular dynamics (MD) simulations (Figure 2A) by measuring current blockages (in experiments and simulations) and dwell times (from experiments only) for MC-LR (and MC-RR, -YR in Figure S12) in WT AeL and comparing them with two mutants that alleviate the two constriction points, K238A and R220A (Figure 2B and 2C). In the simulations, the MC-LR molecule was placed inside the β -barrel, in between the two constriction points, near K238—a conformation we hypothesized is dominant for producing MC-LR current blockages (Figure 2A)—spanning residues from S228/Q268 to G234/N262 (residue numbers correspond to the two β -strands of AeL's barrel) and with no MC atoms within 4 Å of any protein atoms. In both experiments and simulations, currents were measured in symmetric 4 M KCl buffer.

Experimentally, MC-LR induced comparable levels of blockage in both WT and R220A pores, although the dwell times differed between the two pores (Figure 2B). In contrast, the K238A mutant exhibited significantly reduced blockage compared to WT and R220A but a similar dwell time as the WT AeL pore. On the basis of these findings, the main current blockage (sensitivity) seems modulated only by the K238 constriction, as mutation of the R220 constriction does not alter the current compared to WT. This is further supported by MD simulations, which show a similar trend in blockage percentages across the three different pores (WT, K238A, and R220A; Figure 2C) when MC-LR was positioned within the pore cavity as stated above. In contrast, dwell time seems to be

primarily regulated by the R220 constriction, since the R220A mutant exhibits a shorter dwell time for MC-LR while maintaining the same current blockage as that of WT.

Expanding the Detection Range of Microcystins

To further test the ability of WT AeL to discriminate MCs, the electrical readouts of four additional MC congeners were recorded. These congeners possess a fixed leucine residue at position X (Figure 1B) and a variable residue at position Z, being alanine, phenylalanine, tyrosine, or tryptophan, depicted in Figure 3 as MC-LA, -LF, -LY, and -LW, respectively. At 100 mV, MC-LF, MC-LY, and MC-LW exhibited extremely long dwell times, resulting in very low event frequencies for reliable current histograms. We therefore increased the voltage to 160 mV, where MC-LR, MC-YR, and MC-RR remained discernible, enabling also confident detection for the more hydrophobic variants (MC-LA, MC-LY, MC-LW, and MC-LF), as shown in Figure 3. Under a voltage of 160 mV, we were able to separate all the tested MC congeners, as shown by the current blockage histograms obtained from individual experiments (Figures 3A, S16). More hydrophobic MCs, especially MC-LF and -LW, produced significantly longer dwell times at 160 mV compared to other MCs (Figure S15). The dynamic range of our method is reliably achievable up to 50 nM for the tested microcystins, except for MC-LY and -LW, for which optimal signal can only be observed at concentrations up to 25 nM. This range is suitable for detecting environmentally relevant concentrations, as MC levels during cyanobacterial blooms rarely exceed 50 nM.^{13,80–86} MC-LY and -LW required

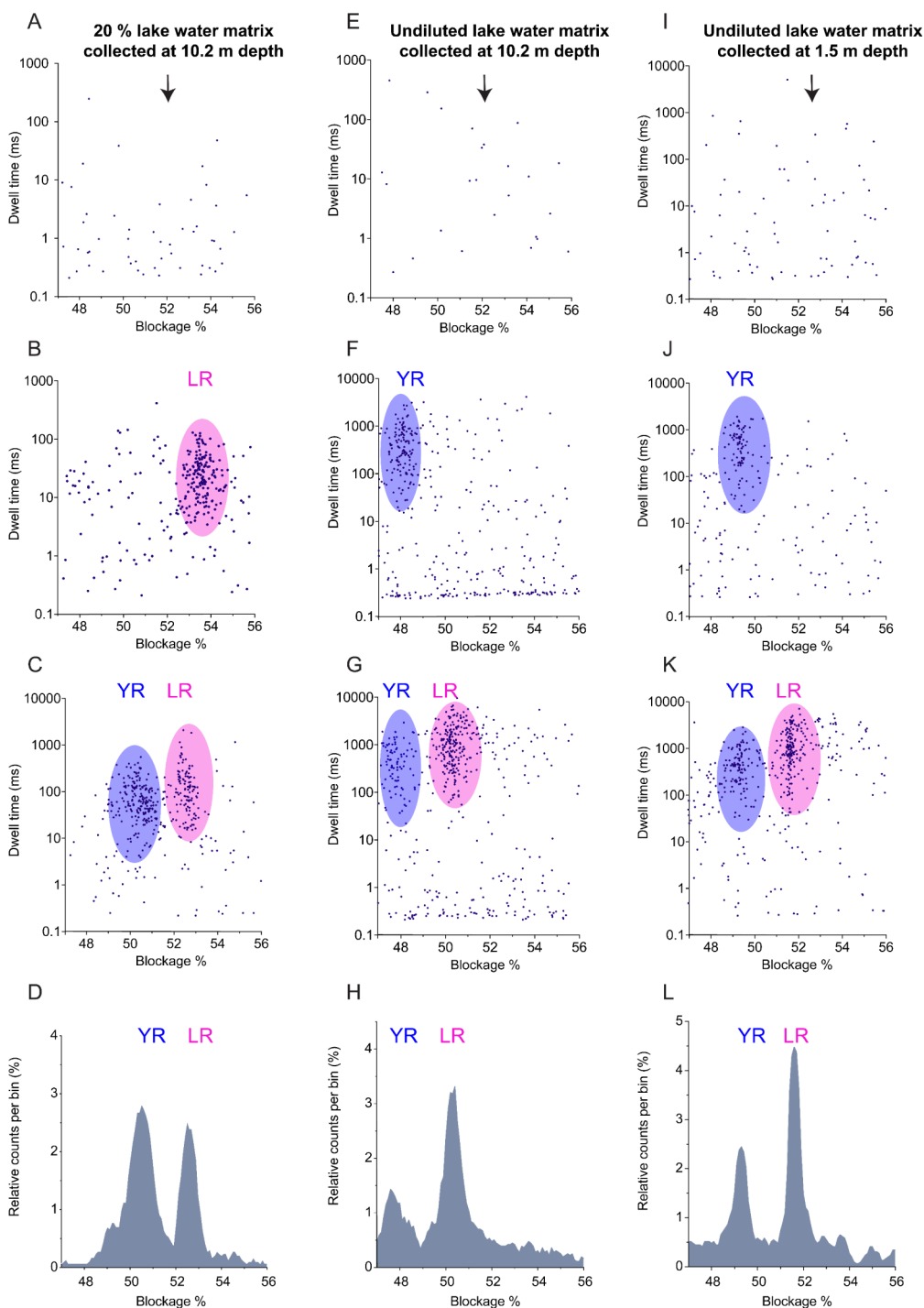


Figure 4. Discrimination of MCs in a lake water matrix. (A–D) Top to bottom: Diluted matrix containing 20% lake water collected at 10.2 m depth, the scatterplots of the single pore upon sequential addition of MCs to the matrix, histogram (bin size = 0.1) of the mixture experiment in C, respectively. (E–H) Undiluted lake water background collected at 10.2 m depth, the scatterplots of the single pore upon sequential addition of MCs to the matrix, histogram (bin size = 0.1) of the mixture experiment in G, respectively. (I–L) Undiluted lake water background collected at 1.5 m depth, the scatterplots of the single pore upon sequential addition of MCs to the matrix, histogram (bin size = 0.1) of the mixture experiment in K, respectively. Data shown were recorded at 100 mV, 3.3 M KCl for A–D, and 4 M KCl for E–L.

testing at 25 nM because, at higher concentrations (e.g., 50 nM), they frequently produced two-level blockade events (Figure S17) that hindered rapid and confident identification. These events often failed to return to the open-pore current, particularly at lower voltages (e.g., 100 mV). A similar behavior was initially observed for MC-LR at 2 μ M (Figure S18), which was resolved by lowering the analyte concentration. Based on

these observations, we propose two hypotheses to explain the two-level events observed with MC-LY and -LW. First, they may result from strong aromatic interactions (e.g., π – π or cation– π) between the MCs and the pore,⁸⁷ resulting in MC getting stuck in 2 possible states within the pore. Second, two MC-LY or -LW (and similarly -LR at 2 μ M) molecules may be entering the pore sequentially or in a chain-like manner,

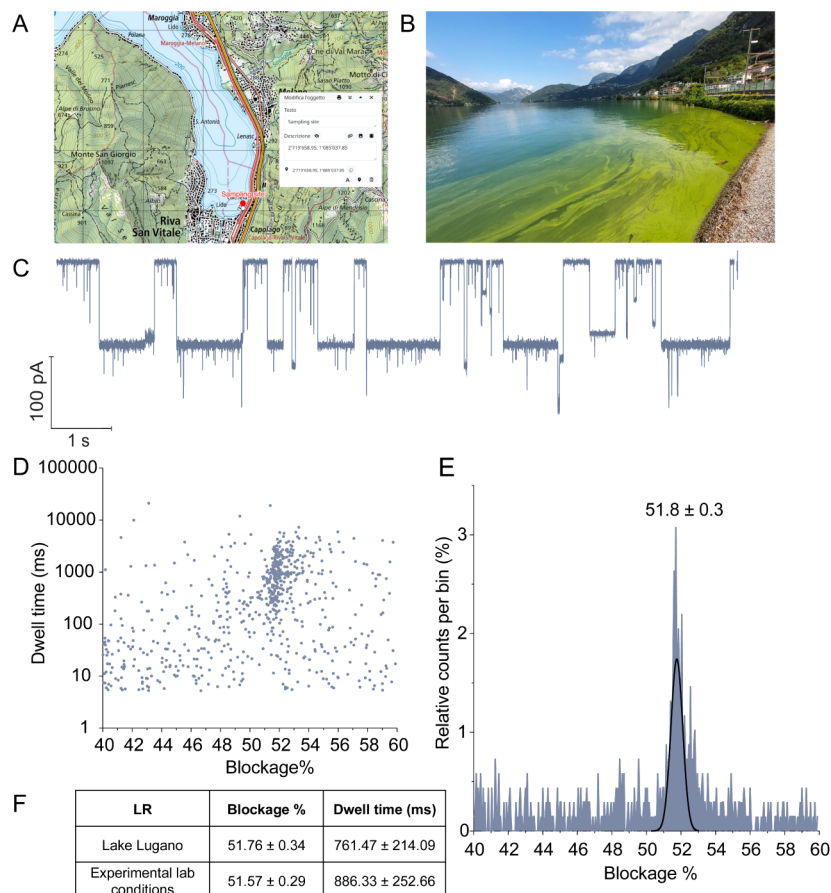


Figure 5. Detection of MC-LR from Lake Lugano with WT AeL nanopore. (A) Location map of the sampling site at Lake Lugano (© Data: swisstopo). (B) Strong cyanobacterial bloom from which the sample was collected. Photo courtesy of Centro di competenze sull'acqua (CCA) SA, Chiasso (CH). (C) Typical trace recorded at 100 mV in 4 M KCl buffered with 10 mM Tris, 1.0 mM EDTA at pH 7.5. Selected 10 s fragment of the trace, recorded at a sampling rate of 50 kHz and filtered with a 500 Hz low-pass filter. (D) Overlay of scatterplots across at least $n = 3$ pore replicates produced by MC-LR in Lake Lugano water interaction with WT AeL, event selection region (40–60%) is visualized (full range shown in Figure S23). (E) The histogram from current in B. (F) Table comparing current blockage % and dwell time of MC-LR from Lake Lugano (individual dwell time fits per pore in Figure S23) and the experimental lab conditions, where the average and standard deviation across $n = 3$ replicates are shown.

potentially driven by intermolecular aromatic interactions that increase affinity and facilitate cooperative entry and exit dynamics.

Bulkier MCs such as MC-LF, -LY, and -LW provided deeper pore blockage than -LA, seemingly the smallest MC based on structure. However, MC-LA produced deeper blockage than -RR (Figure 3A), probably due to its higher hydrophobicity compared to -RR, as previously discussed. Separation of the 7 MC species was clearly evident in dwell time and current blockage dimensions, without the need for further sophisticated analysis (Figure 3B). We observed that all the MCs did possess a similar dwell time for this voltage except MC-LA, probably due to its smaller size. The results demonstrate that the main parameter to discriminate MC is current blockage, since very narrow current distribution can be observed. To go a step beyond, a mixture experiment with all 7 MC congeners present in a single sample was performed, which is relevant for real-world applications (Figure 3C and Figure S27). As shown, the individual current slightly shifted, but the specific population corresponding to each MC remained conserved and did not overlap, demonstrating the capabilities of WT AeL to discriminate up to seven MC congeners simultaneously. We used an LSTM classifier operating on segmented 14-dimen-

sional time–frequency feature vectors (amplitude statistics, higher-order moments, and FFT-based spectral descriptors) extracted from each event trace to assign every blockade to its corresponding MC congener, achieving >85% per-class accuracy for the first three congeners and >70% for most congeners when all seven were included (Figure S24). These results highlight the WT AeL nanopore's discriminatory power for multiplexed MC detection in complex mixtures.

Discrimination of Microcystins Spiked on Lake Water Samples

Discrimination of up to 7 MCs was achieved in lab-controlled conditions; subsequently, we focused on assessing MC detection in natural surface water. For that, we assessed the detection of MC-LR and -YR in water sourced from Lake Geneva, Switzerland (Figure 4). This condition was anticipated to pose a challenge due to the minimal difference in current blockage (2.1% from Figure 1D). Lake water without detectable MC (as determined by ELISA in Figure S19; see Methods) was used as a background matrix. MC-LR and MC-YR were spiked at 25 nM and 20 nM, respectively—concentrations resembling those found in lake environments affected by strong cyanobacterial blooms.⁴⁵ Lake water background signals were recorded in unspiked samples. As

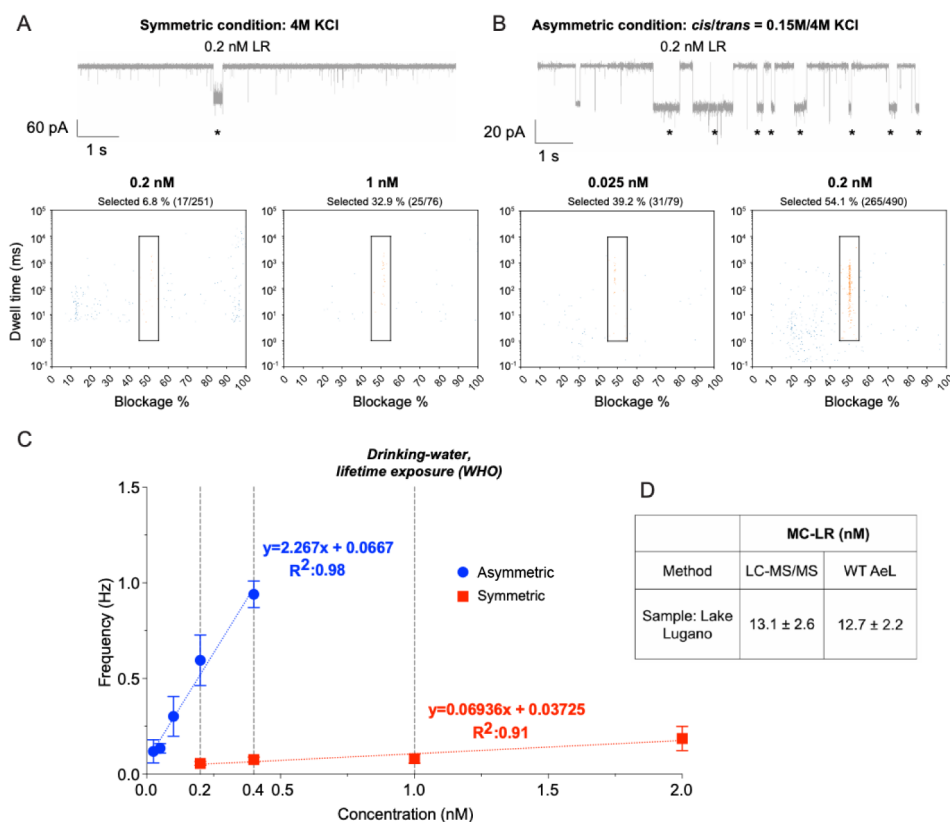


Figure 6. Probing the microcystin detection limits. (A) Raw current trace, where MC-LR events are marked with a star, and scatterplots in symmetric conditions at 0.2 nM and 1 nM MC-LR. (B) Raw current trace, where MC-LR events are marked with a star, and scatterplots in asymmetric conditions at 0.025 nM and 0.2 nM MC-LR. (C) Frequency (Hz) of MC-LR events selected from scatterplots in symmetric (red) and asymmetric conditions (blue). Data acquired at 100 mV in the respective KCl concentrations, buffered with 10 mM Tris, 1.0 mM EDTA at pH 7.5. The strictest WHO guideline value for lifetime exposure to MC-LR in drinking water, 1 nM, is shown above. Each point is the result of the average of at least $n = 3$ pore replicates. (D) The event frequency of MC-LR in Lake Lugano yielded the concentration of 12.7 nM, close to the LC-MS/MS value of 13.1 nM.

shown in the scatterplots in Figure 4A, E, I, lake water did not produce any significant background, and mainly short events could be observed. More importantly, there were no events within the current blockage windows expected for MCs, which could otherwise interfere with MC discrimination.

First, MC-LR and -YR were spiked into a diluted lake water matrix (at 3.3 M KCl instead of the 4 M KCl previously used; see Methods for water sample preparation, Figure S20A) and the same resolution in current blockage was retained as observed in Figure 1G, with a separation of approximately 2.5% between MC-YR and -LR (Figures 4A–D, S20D). Next, we directly used water collected at 10.2 m (Figures 4E, S20B) and 1.5 m depths (Figures 4I, S20C), representing water from the deep chlorophyll maximum (DCM) layer and the surface layer, respectively. To keep similar experimental conditions and assess the background noise resulting from any other (in)organic material present, the same electrolyte concentration (4 M KCl) was kept for the following experiments, and the MC-YR and -LR were spiked sequentially (Figures 4F–H, S20E, 4J–L, S20F). Assessing this under high-salt conditions is necessary, as elevated salt levels can enhance the capture of any molecules from lake water, potentially contributing to signals that interfere with MC detection. Sequential spiking of the MCs to the respective samples demonstrated that the current separation was preserved (Figure 4H, L). As shown in Figure 4D, H, L, the current separation was still preserved as in individual and mixture experiments performed previously

(Figure 1D). A slight shift in current is observed, which can be attributed to the different ion composition present in the lake water, but with no apparent consequence for the separation of MC-LR and -YR ($2.4 \pm 0.2\%$ compared to 2.1% in Figure 1D). These results demonstrate that the AeL nanopore enables reliable and selective detection of these microcystin congeners in natural water samples, at environmentally relevant concentrations during blooms, without interference from the sample matrix.

Microcystin Detection from Contaminated Lake Water Samples

To demonstrate the feasibility of nanopore-based detection for microcystins in natural water containing endogenously produced toxins, a surface water sample was collected from Lake Lugano (2 m from shore) during a cyanobacterial bloom (Methods, Figure 5A, B, Figure S21). Prior to nanopore analysis, LC-MS/MS confirmed the presence and quantified the amount of MC-LR at $13.038 \mu\text{g/L}$ (13.1 ± 2.6 nM) (Figure S22, Figure S26, Table S5). Following preparation as described in the Methods, the sample produced recognizable MC-LR-specific current blockades in nanopore experiments, as shown by the current trace, scatterplot, and blockage histogram (Figure 5C–E, Figure S23). MC-LR from Lake Lugano yielded similar blockage percentage and dwell time (ms) values as MC-LR in pure experimental lab conditions (Figure 5D), demonstrating the robustness of the method developed here.

Pushing the Microcystin Detection Limit to Reach Picomolar Concentrations

To further enhance the sensitivity of the nanopore-based detection system, a series of experiments was performed to evaluate the concentration-dependent response and establish the linear detection range for MC-LR.³¹ Our aim is to detect concentrations well below the WHO's lowest guideline value of 1 nM for chronic lifetime exposure in drinking water,⁸⁸ as well as environmentally relevant concentrations typically found in lake surface waters during bloom events.¹⁰ In concentrated cyanobacterial scums, MC levels can even reach micromolar concentrations (up to 14000 $\mu\text{g/L}$ (14 μM) of total MC).⁸⁹ However, in open surface or lake waters, MCs are usually reported in the nM range or lower (ng/L to low $\mu\text{g/L}$). For example, microcystin levels can peak between 105 pM and 147 pM¹³ in Lake Geneva to approximately 1 nM (1,000 ng/L) in other Swiss lakes,^{80,81} 10 nM in highly eutrophic lakes in the USA,⁸² 24 nM in the southern US Reservoir,⁸³ 46 nM in Hoan Kiem Lake (Vietnam),⁸⁴ 50 nM in Lake Taihu (China),⁸⁵ and up to 78 nM in lakes in Mexico.⁸⁶ To improve the sensitivity, we employed a salt concentration gradient, a technique previously demonstrated to increase event frequency for ssDNA.⁹⁰ Specifically, we used asymmetric salt conditions with a 0.15 M KCl buffer solution in the *cis* chamber and a 4.0 M KCl buffer solution in the *trans* chamber (Figure 6).

Figure 6 shows two representative raw current traces for 200 pM MC-LR under symmetric (A) and asymmetric conditions (B). The number of events (depicted by stars) is drastically improved under asymmetric conditions. This is further demonstrated in the scatter plots, where MC-LR events (highlighted within a dark rectangle) account for 54.1% of total events under asymmetric conditions, compared to only 6.8% under symmetric conditions.

Event frequency (in Hz) was calculated by fitting interevent time (Figure S15 and Figure S28), where concentrations of MC-LR ranging from 25 pM to 2 nM were screened under both symmetric and asymmetric conditions, and the frequency of events corresponding to microcystin was extracted and plotted (Figure 6C; Table S4). Based on the scatterplots (Figure 6B) and the event frequency (Figure 6C), under symmetric conditions, MC-LR was confidently detected at concentrations as low as 1 nM, accounting for 32.9% of observed events and yielding 0.081 ± 0.02 Hz. At a concentration of 0.2 nM, the frequency of events under symmetric conditions was $0.055 \text{ Hz} \pm 0.01$ and under asymmetric conditions was $0.595 \text{ Hz} \pm 0.1$, resulting in a 10.7-fold improvement. Similarly, an improvement of the event frequency at 0.4 nM was observed under asymmetric conditions with a 12.4-fold improvement. Using this approach, we can therefore significantly improve the detection capability of WT AeL, enabling reliable MC-LR detection at concentrations substantially lower than the 1 nM WHO guideline. This sensitivity is well below the concentrations for water safety and environmental monitoring, demonstrating the strong potential of this method for real-world applications. Finally, using the standard curve obtained for symmetric conditions, we quantified the MC-LR concentration in Lake Lugano, which yielded a concentration of 12.7 ± 2.2 nM. This concentration corresponds well to the one measured by LC-MS/MS (13.1 ± 2.6 nM), confirming that aerolysin nanopores enable both identification and quantification of microcystin concentrations in environmental lake water samples.

CONCLUSIONS

Using the aerolysin nanopore, we demonstrate real-time detection of 7 MC congeners with high sensitivity and specificity, competing with the state-of-the-art LC-MS/MS technique. We successfully detected the congeners at nanomolar concentrations, demonstrated the feasibility to use and quantify microcystin-LR in contaminated lake water, and achieved picomolar detection under asymmetric buffer conditions. This represents a significant advancement over previous studies and positions nanopore-based detection as an effective technology for real-world applications in water surveillance. The method's ability to discriminate seven MC congeners within mixture experiments, including those spiked in lake water, as well as the identification and quantification of MC-LR in naturally contaminated lake water, underscores the reliability and robustness of the aerolysin nanopore platform for direct, real-time monitoring of microcystins.

Beyond microcystins, water sources worldwide are threatened by a diverse array of contaminants, including other cyanotoxins (such as anabaenopeptins, cylindrospermopsins, anatoxins, or saxitoxins), pesticides, pharmaceuticals, heavy metals, or other industrial pollutants.^{14,15,91} The present study highlights that the aerolysin nanopore can, in principle, be used as an analytical tool for the detection of these water contaminants. Through protein engineering or chemical modifications, aerolysin's sensitivity and selectivity can be further expanded to cover sensing of a wider range of (cyano)toxins or small molecules, as shown in other domains for nanopores such as MspA or α -hemolysin.⁹² Moreover, the adaptability and compact nature of the nanopore technology make it ideal for integration into portable, field-deployable devices for on-site water quality monitoring.^{93–95}

Lastly, the ability to rapidly and accurately monitor multiple toxins in real-time is crucial for safeguarding water quality and protecting public health. Our results support the development of more comprehensive and responsive water quality management strategies, addressing critical environmental and public health concerns associated with cyanotoxin and other pollutant contamination.

METHODS

Preparation of Aerolysin

The wild-type aerolysin full-length sequence was cloned into the pET22b vector with a C-terminal hexahistidine tag as in a previous study.⁷⁵ For the preparation of K238A and R220A, site-directed mutagenesis on the aerolysin gene was carried out using the QuikChange II XL kit (Agilent Technologies) according to the manufacturer's protocol, followed by expression and purification⁵⁴ from *Escherichia coli* BL21 DE2 pLys cells. The bacterial culture was cultivated in Luria-Bertani medium until it reached an optical density between 0.6 and 0.7. Protein expression was then induced with 1 mM isopropyl β -D-1-thiogalactopyranoside (IPTG), followed by overnight incubation at 20 °C. The harvested cells were suspended in a buffer containing 20 mM sodium phosphate (pH 7.4) and 500 mM NaCl and mixed with cOmplete Protease Inhibitor Cocktail (Roche). Cells were lysed by sonication, after which the lysate was centrifuged at $20,000 \times g$ for 35 min at 4 °C. The clarified supernatant was loaded onto a pre-equilibrated HisTrap HP column (GE Healthcare) in the same buffer. Protein elution was performed using a 40-column volume gradient of buffer containing 20 mM sodium phosphate (pH 7.4), 500 mM NaCl, and 500 mM imidazole. Subsequently, the eluted protein underwent buffer exchange into a final solution of 20 mM Tris and 150 mM NaCl (pH 7.4) using a HiPrep Desalting column (GE

Healthcare). The purified protein was then rapidly frozen in liquid nitrogen and preserved at -20°C for future use.

Nanopore Single-Channel Measurements

Powdered 1,2-Diphytanoyl-*sn*-glycero-3-phosphocholine (DPhPC) lipid (Avanti Polar Lipids, Inc., Alabaster, AL, USA) was dissolved in octane (Sigma-Aldrich Chemie GmbH, Buchs, Switzerland) to get a final concentration of 8 mg/mL. The purified protein was diluted to a concentration of 0.2 $\mu\text{g/mL}$ and activated using trypsin-agarose (Sigma-Aldrich Chemie GmbH, Buchs, SG, Switzerland) for 2 h at 4°C . Subsequently, the solution was centrifuged (11,000 rpm, 5 min, 4°C) to remove trypsin.

Single-channel current recording experiments were performed on an Orbit Mini workstation (Nanonion, Munich, Germany), using MECA 4 recording chambers with a 50 μm cavity size. DPhPC membranes were formed across the four cavities of the chamber, separating the well into *cis* and *trans* compartments. Each microcavity contains an integrated Ag/AgCl microelectrode through which an electrical bias is applied. The heptameric pore is formed upon addition of aerolysin to the *cis* chamber, followed by its self-assembly and insertion into the membrane. The experiments were performed at 25°C under symmetrical buffer conditions with both compartments filled with a 150 μL electrolyte solution, where MC analytes were added to the *cis* compartment. All data was gathered using a sampling rate of 100 kHz and filtered with a 5 kHz low-pass filter. In order to obtain relatively uniform capture between MCs, the ratio of MC-LR:RR:YR used for the mixture was chosen based on the capture rate extracted from individual MC experiments (Figure S4). Since the ratio of MCs expected to result in similar capture was varying depending on voltage, that of 160 mV was chosen.

To perform asymmetric salt experiments, Teflon films with 50 μm diameter openings were secured between two compartments of a Teflon chamber using high-vacuum grease (Dow Corning Corporation, Midland, MI, USA). *Cis* and *trans* compartments were connected solely through the aperture in the film, with each side containing an Ag/AgCl electrode. Both sides of each aperture were pretreated with 1 μL of a 1% (v/v) hexadecane (Sigma-Aldrich Chemie GmbH, Buchs, Switzerland) solution in hexane (Sigma-Aldrich Chemie GmbH, Buchs, Switzerland), after which the chamber was placed in the recording apparatus. DPhPC bilayers were then established across the aperture using the folding technique described in earlier protocols.^{96,97} Electrolyte solution was introduced into both chambers (70 μL of 0.15 M KCl in *cis* and 600 μL of 4 M KCl in *trans*), ensuring that the liquid remained below the level of the aperture. Lipids, dissolved at 10 mg/mL in pentane (Sigma-Aldrich Chemie GmbH, Buchs, Switzerland), were then applied to the electrolyte surface in each compartment. Once the pentane evaporated, the electrolyte volume was raised above the aperture by adding 110 μL of 0.15 M KCl to *cis* and 800 μL of 4 M KCl to *trans* to facilitate bilayer formation. Upon adding WT AeL in small quantities and subsequent pore formation, after a few minutes of equilibration, the pipet offset voltage knob on the Axopatch was applied to cancel the baseline offset, establishing a 0 pA baseline at 0 mV. The integrity of the resulting lipid bilayer was assessed by measuring its capacitance. When an analyte was added, the *cis* compartment was mixed by pipetting to ensure thorough distribution. Electrical currents were recorded at a sampling rate and a low-pass filter of 100 kHz using an Axopatch 200B amplifier (Molecular Devices, LLC, San Jose, CA, USA).

Current Signal Processing

Information regarding open pore current (I_0), mean residual current (I), average relative current (I/I_0), event dwell time (dwt), and interevent time were extracted from the raw current traces and processed with an in-house developed Python-based pipeline. Current blockage was calculated as the difference between the open pore current (I_0) and the current produced by the passage of the analyte through the pore (I), expressed as $\Delta I/I_0$ (where $\Delta I = I_0 - I$), and reported as "Blockage %" in the figures. The raw signals are divided according to voltage discontinuities and significant time-scale shifts to identify stable single pore readings and discard segments where the

pore self-blocks or where several pores are present. For each identified segment, the distribution of the open pore current is analyzed by applying a Gaussian fit to the peak of the current distribution that exhibits the highest mean value. Event detection is performed using a dual-threshold approach based on the open-pore current distribution: 1) for event initiation, a threshold of 7σ below the mean open-pore current is applied, where σ represents the standard deviation of the open-pore current; 2) for event termination, a threshold of 3σ below the mean open-pore current is applied. The average relative current is translated into the pore blockage % ($\Delta I/I_0$). The processed data is initially visualized as scatterplots, enabling the selection of events of interest (falling in the blockage range of 40–60% and dwt range of 0.2 ms–10 s) for further in-depth analysis, where both current and dwell time distributions are identified by fitting a Gaussian and an Exponential function, respectively. The event selection window of 40–60% blockage was chosen based on the range where specific and reproducible events corresponding to the respective MC congeners consistently appeared, forming a well-defined population within this interval. Occasional events observed at lower or higher blockage levels were not consistently present across different pores and are attributed to transient bumping interactions or increased background noise under 4 M KCl conditions (lower blockage), as well as occasional pore clogging or self-blockage events (higher blockage). MCs produced rather defined current blockage distributions, with more spread dwell time distributions; the event selection windows of the respective MCs are shown in Figures S5–S8, S14. For event frequency analysis, the selection window was modified (to a 45–55% current blockage and 1 ms–10 s dwell time range) to more accurately detect MC-LR events, and the resulting frequency of events was fitted with a single exponential as shown in Figure S28 for both symmetric and asymmetric conditions for different concentrations.

Lake Water Collection and Sample Preparation

Lake water samples of Lake Geneva (Switzerland) were collected on September 16th, 2022, from the LEXPLORÉ⁹⁸ experimental platform. Samples were collected at the depths of 1.5 and 10.2 m below the surface. Prior to nanopore experiments, all samples were filtered through a 0.2 μm membrane to remove particles, intact bacteria, and other microorganisms. To validate the suitability of the matrices as negative controls for microcystin detection, an Enzyme-Linked Immunosorbent Assay (Biorbyt, UK) was performed to confirm the absence of microcystins per the manufacturer's instructions. The assay is a direct competitive (dcELISA) format: MC toxin in the sample competes with horseradish peroxidase-conjugated MC for a limited amount of antibody coated in the wells. Color development was read as OD at 450 nm using an ELISA reader. Measurements were performed in duplicate (Figure S19). OD of each set of calibrators (C1→C5) and samples were averaged, and the normalized mean OD value was calculated as follows: average OD of calibrator or sample/average OD of negative control (NC). A standard curve of normalized mean OD vs MC concentration was plotted, and sample MC concentrations were obtained by interpolation. The diluted lake water matrix (20% v/v) shown in Figure 4A–D was prepared by adding 30 μL of lake water to 120 μL of 4 M KCl solution, resulting in a final KCl concentration of 3.3 M (total background recording time: 10 min; total spiked MC recording time: 14 min). The undiluted lake water matrix shown in Figure 4E–H, I–L was prepared by directly adding solid KCl crystals to the 0.2 μm filtered water samples collected at 10.2 m (total background recording time: 30 min; total spiked MC recording time: 20 min) or 1.5 m (total background recording time: 20 min; total spiked MC recording time: 20 min) to reach the final concentration of 4 M KCl. After thoroughly mixing lake water samples with KCl crystals, the mixture was transferred to MECA chips, where the lipid bilayer was formed and aerolysin nanopore added.

A Lake Lugano sample was collected on September 15th, 2025, in the Capolago/Mendrisio area (coordinates: 2°19'658.95, 1°08'037.85). The sample was obtained from Centro di competenza sull'acqua, CCA SA, Chiasso. It was collected from the surface, at 2 m from the shoreline, as shown in Figure S21, and prepared as

previously explained by filtration and direct addition and mixing with salt crystals, yielding a KCl concentration of 4 M.

LC–MS/MS

The Lake Lugano sample was analyzed for the most common microcystins (MC-RR, MC-YR, and MC-LR) using LC–MS/MS. Calibration standards for these analytes were prepared from reference materials purchased from Sigma-Aldrich. A stock solution of a mixture of these three microcystins was prepared at a concentration of 5 µg/mL. Calibration solutions, with concentrations of 0.025 µg/L to 0.500 µg/L, were prepared by serial dilution of the stock solution. Nodularin was used as an internal standard. LC–MS/MS analysis was performed on an ACQUITY Class I system from Waters coupled to a TQ-S micro from Waters.

Liquid chromatography (LC) analyses were performed using an Acquity UPLC HSS T3 analytical column (1.8 µm, 2.1 × 150 mm; Waters). Mobile phase A consisted of water with 0.04% acetic acid, and mobile phase B was methanol. The column temperature was maintained at 50 °C, with a sample injection volume of 250 µL and a flow rate of 0.34 mL/min. The LC gradient profile is detailed in Table S6.

Mass spectrometry (MS) analyses were conducted in the positive electrospray ionization (ESI) mode. Multiple reaction monitoring (MRM) transitions are specified in Table S7.

Quantitative analysis showed excellent linearity in detector response from 0.025 µg/L to 0.500 µg/L for MC-RR, MC-YR, and MC-LR. Calibration curves (Figure S22) yielded coefficients of determination (R^2) of 0.9963, 0.9951, and 0.9959, respectively. The limit of quantification (LOQ) was 0.025 µg/L for each microcystin, enabling detection of MC-LR well below the WHO guideline of 1 µg/L. LC–MS/MS analysis of the Lake Lugano sample with the chromatogram and spectrum is shown in Figure S26.

NMR Spectroscopy

All NMR experiments were carried out in a Bruker 800 MHz spectrometer equipped with an Avance Neo console and a triple-resonance cryoprobe at 298 K. Samples were prepared in 20 mM phosphate (pH 5.9) buffer⁷⁵ at concentrations close to 1 mM (MC-LR and MC-RR) or 0.1 mM (MC-YR). Resonances were assigned by combining ¹H,¹H TOCSY, ¹H,¹H NOESY, and ¹H,¹³C HSQC experiments (the latter only for MC-LR and MC-RR), guided by assignments available in the literature for MC-LR.^{75,99,100}

MD Simulations

We set up the systems for MD simulations by using our recently published high-resolution cryo-EM structure of WT AeL, noting in particular several key protonation states critical for proper simulation as described in the paper reporting said structure along with MD simulations.⁵¹ For this work, we parametrized the systems and carried out the simulations by using the same CHARMM-GUI-based protocol, utilizing the same DPhPC membrane but with a solution of 4 M KCl as employed in the experiments. The MC molecules were parametrized with the CHARMM General Force Field.¹⁰¹ The simulations were run with Gromacs 2024 by using the standard production stage scripts produced by CHARMM-GUI (pressure coupling to 1 atm, a temperature of 303 K, 2 fs integration steps, LINCS-based restraints on hydrogens, and PME electrostatics with a 12 Å cutoff), but applying the indicated voltages along the z dimension after the standard minimization and equilibration protocol obtained from CHARMM-GUI. The simulations were visualized with VMD and processed with custom scripts to obtain currents by counting the numbers of K⁺ and Cl[−] ions passing per unit time, from which the blockages were then calculated. To validate the force field used, we computed the RDF from K⁺ to Cl[−] ions in 1 M KCl (which has been used before)^{51,102} and 4 M KCl simulations. The results (Figure S25) show no significant difference between the two simulations, indicating no artificial aggregation due to any saturation. Moreover, the simulation at 4 M, 100 mV showed a current of 262.95 pA, close to experimental data.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.6c05413>.

Reagents, microcystin standards used, and stock solutions prepared; **Supplementary Tables S1–S7** include information regarding WHO guideline values, physical and chemical properties of the tested MC congeners, chemical shifts data, event frequency analysis, and LC–MS/MS details; **Figures S1–S28** include raw current traces, scatterplots, blockage % and dwell time histograms, NMR spectra, LC–MS/MS calibration curves, LSTM confusion matrix, MD details, and event frequency histograms (PDF)

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Author Contributions

A.A., T.K., and M.D.P. conceived the project; A.A., J.F.B.J., T.K., and M.D.P. wrote the original draft; all authors contributed to the final manuscript; A.A. and J.F.B.J. performed and analyzed nanopore experiments; M.J.M. produced aerolysin pores; L.A.A. performed and analyzed NMR experiments and MD simulations; L.W.P. performed and analyzed MD data; C.C. provided guidance in the analysis of the nanopore data; A.C. collected the lake water samples; E.M.L.J. advised on the handling of microcystins; A.C. and E.M.L.J. provided insights and contributed to the interpretation of the environmental aspects of the project; G.V., N.S., S.B., and F.K. provided the LC-MS/MS and ELISA data; T.K. and M.D.P. supervised the project and acquired funding.

Notes

A preprint version of this article is available on the bioRxiv repository.¹⁰³
The authors declare no competing financial interest.

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